The extension of Ballantine and Merca's conjecture on 6-regular partitions

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Abstract. In 2012, Andrews and Merca gave a truncated version of Euler's pentagonal number theorem which leads to a recurrence relation for partition functions. This work has inspired the subsequent research on truncated theta series. In 2021, Ballantine and Merca proposed a conjecture on 6-regular partitions when they study the truncated pentagonal number series. In this paper, by considering a truncated theta series involving ℓ-regular partitions, we confirm Ballantine and Merca's conjecture on 6-regular partitions and further show that it still holds for $k \geq 1$ when $\ell = 3$ or $\ell \geq 5$. Moreover, it also confirms Merca's conjecture on partitions with parts not congruent to $0, \pm 3 \pmod{12}$ and Merca's stronger conjecture on the truncated Jacobi triple product series with $R = 3S$ and $S \geq 1$.

Keywords: truncated theta series; ℓ-regular partitions; Jacobi's triple product series; Euler's pentagonal number theorem

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1. Introduction

It is well known that Euler's pentagonal number theorem [1, p.11]

$$(q;q)_\infty = \sum_{n=0}^{\infty} (-1)^n q^{n(3n+1)/2}(1-q^{2n+1}) \tag{1.1}$$

is one of the most famous identities and plays an important role in the theory of partitions and q-series. It is closely related to the partition function $p(n)$ [1] which counts the number of partitions of n and has the following generating function

$$\sum_{n=0}^{\infty} p(n)q^n = \frac{1}{(q;q)_\infty}. \tag{1.2}$$

Thereafter, Euler's pentagonal number theorem leads to the recurrence relation for $p(n)$ as follows

$$p(n) + \sum_{j\geq 1} (-1)^j (p(n-j(3j-1)/2) + p(n-j(3j+1)/2)) = 0.$$

In the field of the partition theory, the inequalities for various types of partition functions has been studied extensively. It is easy to see the initial relation $p(n) - p(n-1) > 0$, while such relations with more terms are more meaningful. In [10], Merca

stumbled across the following inequality for the partition function

$$p(n)-p(n-1)-p(n-2)+p(n-5)\leq 0. \tag{1.3}$$

In 2012, Andrews and Merca [2, Lemma 1.2] used Shanks' truncated version [14, (2)] of Euler's pentagonal number theorem, that is,

$$\sum_{j=-k}^{k}(-1)^j q^{j(3j+1)/2}=\sum_{n=0}^{k}\frac{(-1)^n(q;q)_k q^{\binom{n}{2}+(k+1)n}}{(q;q)_n},$$

to derive the truncated identity

$$\frac{1}{(q;q)_\infty}\sum_{j=-k}^{k-1}(-1)^j q^{j(3j+1)/2}=1+(-1)^{k-1}\sum_{n=1}^{\infty}\frac{q^{\binom{k}{2}+(k+1)n}}{(q;q)_n}\begin{bmatrix}n-1\\k-1\end{bmatrix}. \tag{1.4}$$

From the above result, they derived that for $k\geq 1$,

$$(-1)^{k-1}\sum_{j=0}^{k-1}(-1)^j\big(p(n-j(3j+1)/2)-p(n-j(3j+5)/2-1)\big)\geq 0 \tag{1.5}$$

with strict inequality if $n\geq k(3k+1)/2$. They also gave a combinatorial interpretation for the above expression [2, Corollary 1.3].

In 2013, Guo and Zeng [7] derived a family of inequalities for the overpartition function $\overline{p}(n)$ and for the partition function $\mathrm{pod}(n)$ which is the number of partitions of n with distinct odd parts. In the study of truncated theta series, Guo and Zeng [7] and Andrews and Merca [2] proposed the following conjecture for truncated Jacobi triple product series, independently.

Conjecture 1.1. *For positive integers ℓ, R, S with $1\leq S<R/2$, the coefficient of q^n with $n\geq 1$ in*

$$(-1)^{\ell-1}\frac{\sum_{j=0}^{\ell-1}(-1)^j q^{R\frac{j(j+1)}{2}-Sj}\left(1-q^{(2j+1)S}\right)}{\left(q^S,q^{R-S},q^R;q^R\right)_\infty} \tag{1.6}$$

is nonnegative.

In 2015, Mao [9] and Yee [17] proved Conjecture 1.1 algebraically and combinatorially, respectively. Zhou [18] confirmed Conjecture 1.1 in seven special cases for coprime integers α,β,γ by applying the following property from [13].

Proposition 1.2. *Suppose that*

$$\sum_{n\geq 0}R_{\alpha,\beta,\gamma}(n)q^n=\frac{1}{(1-q^\alpha)(1-q^\beta)(1-q^\gamma)}$$

for any distinct integers $\alpha,\beta,\gamma\geq 1$ such that $\gcd(\alpha,\beta)=\gcd(\beta,\gamma)=\gcd(\alpha,\gamma)=1$, then for any $n\geq 0$,

$$R_{\alpha,\beta,\gamma}(n)=\frac{n^2+(\alpha+\beta+\gamma)n}{2\alpha\beta\gamma}+P_{\alpha,\beta,\gamma}(n),$$

where $P_{\alpha,\beta,\gamma}(n)$ is a periodic sequence of period $\alpha\beta\gamma$.

Recently, Merca [11] proposed a stronger version of Conjecture 1.1 as follows.

Conjecture 1.3. *For positive integers $1 \leq S < R$ with $k \geq 1$, the theta series*

$$(-1)^k \frac{\sum_{j=k}^{\infty}(-1)^j q^{Rj(j+1)/2}\left(q^{-Sj}-q^{(j+1)S}\right)}{(q^S, q^{R-S}; q^R)_\infty} \tag{1.7}$$

has nonnegative coefficients.

Ballantine and Feigon [4, Theorem 1.1] proved Conjecture 1.3 combinatorially when $k \in \{1,2,3\}$ and also derived a result which is weaker than the case $S=1$ and $R=3$. Recently, Ding and Sun [6] provided an analytic proof for the special cases $R=3S$ for $S \geq 1$ of the above conjecture.

In 2023, Ballantine and Merca [3] proposed the following conjecture related to 6-regular partitions.

Conjecture 1.4 (Ballantine and Merca [3, Conjecture 2])**.** *For $n \geq 0$, $k \geq 1$,*

$$(-1)^k\Big(\alpha_n - \sum_{j=-k+1}^{k}(-1)^j b_6(n-j(3j-1)/2)\Big) \geq 0$$

with strict inequality if $n \geq k(3k+1)/2$, where $b_6(n)$ is the number of the 6-regular partitions of n, and

$$\alpha_n := \begin{cases}(-1)^m, & \text{if } n = 3m(3m-1), m \in \mathbb{Z}\\ 0, & \text{otherwise.}\end{cases}$$

This conjecture was confirmed by Yao [16] by applying the partition function only with parts $1,2$ or 3 due to Cayley [5]. In this paper, we aim to confirm Ballantine and Merca's conjecture by applying the partitions with non-coprime parts $1,2$, and 4. We also show that Ballantine and Merca's conjecture still holds for more general ℓ-regular partitions with $\ell = 3$ or $\ell \geq 5$.

This paper is organized as follows. In Section 2, we introduce some basic definitions and notations of q-series, and deduce the explicit enumeration formula for partitions with non-coprime parts $1,2$, and 4. In Section 3, we prove Ballantine and Merca's conjecture on 6-regular partitions and further extend it to ℓ-regular partitions with $k \geq 1$, $\ell = 3$ or $\ell \geq 5$. As a consequence, a conjecture proposed by Merca on partitions with parts not congruent to $0, \pm 3 \pmod{12}$ can be confirmed. It also leads to that the stronger Conjecture 1.3 for truncated Jacobi triple product series is true when $R=3S$ with $S \geq 1$.

2. Preliminaries

Throughout this paper, we adopt standard notations and necessary formulas on basic hypergeometric series.

Assume that $|q|<1$ and let the q-shift factorial be defined by

$$(a;q)_n = \begin{cases}1, & n=0;\\ \prod_{j=1}^{n}(1-aq^{j-1}), & n \geq 1,\end{cases}$$

and

$$(a;q)_\infty = \prod_{n=0}^{\infty}(1-aq^n).$$

Recall that a partition of n is a finite nonincreasing sequence of positive integers $(\lambda_1, \lambda_2, \ldots, \lambda_k)$ such that $\lambda_1+\lambda_2+\cdots+\lambda_k = n$. Let $p(n)$ denote the number of integer partitions of n with $p(0) = 1$. As usual, set $p(n) = 0$ if $n < 0$. It is well-known that $p(n)$ has the generating function (1.2).

The ℓ-regular partitions are the partitions with no part is divisible by ℓ, which parameterize the irreducible ℓ-modular representations of the symmetric group when ℓ is prime [8]. Let $b_\ell(n)$ denote the the number of ℓ-regular partitions of n, then its generating function is given by

$$\sum_{n=0}^{\infty} b_\ell(n)q^n = \frac{(q^\ell;q^\ell)_\infty}{(q;q)_\infty}. \tag{2.1}$$

Ballantine and Merca [3] considered 6-regular partitions and the related truncated theta series. By applying Euler's pentagonal number theorem (1.1), they found that

$$(-1)^k\Big(b_\ell(n) - \sum_{j=-k+1}^{k}(-1)^j p\big(n-3j(3j-1)\big)\Big) = \sum_{j=0}^{\lfloor n/6\rfloor} b_\ell(n-6j)M_k(j),$$

where $M_k(n)$ [2, Theorem 1.1] denotes the number of partitions of n in which k is the least integer that is not a part and the number of parts $> k$ is larger than the number of parts $< k$. Furthermore, they derived a recurrence relation for $b_6(n)$ as follows:

$$\sum_{j=-\infty}^{\infty}(-1)^j b_6(n-j(3j-1)/2) = \begin{cases}(-1)^k, & \text{if } n = 3k(3k-1), k\in\mathbb{Z},\\ 0, & \text{otherwise.}\end{cases} \tag{2.2}$$

Inspired by (1.1) and (2.2) , they considered the following series

$$(-1)^k\Big((q^6;q^6)_\infty - \frac{(q^6;q^6)_\infty}{(q;q)_\infty}\sum_{n=-k+1}^{k}(-1)^n q^{n(3n-1)/2}\Big) \tag{2.3}$$

and conjectured that it has nonnegative coefficients. Note that it is equivalent to Conjecture 1.4 by applying Euler's pentagonal number theorem and the generating function for ℓ-regular partitions (2.1) to expand the infinite products and then extracting the coefficient of q^n.

To prove Conjecture 1.4, we need to introduce $p_4(n)$ which denotes the number of partitions of n only with $1, 2$ or 4 as parts, whose generating function is given by

$$\sum_{n=0}^{\infty} p_4(n)q^n = \frac{1}{(1-q)(1-q^2)(1-q^4)}. \tag{2.4}$$

The following expression for $p_4(n)$ also can be found in OEIS [15, A008642]. Here, we provide a combinatorial proof for it.

Lemma 2.1. *For any $n \geq 0$, we have*

$$p_4(n) = \Big(1 + \Big\lfloor \frac{n}{4} \Big\rfloor\Big)\Big(1 + \Big\lfloor \frac{n+2}{4} \Big\rfloor\Big), \tag{2.5}$$

where $\lfloor a \rfloor$ denotes the floor function, i.e., the greatest integer less than or equal to a.

Proof. We prove this lemma by induction. Firstly, it is obvious to see that when $n = 0$ and $n = 1$, $p_4(n) = 1$, and thereby (2.5) holds for $n = 0$ and $n = 1$. Assume that when $n = j$, (2.5) is true, that is,

$$p_4(j) = \left(1 + \left\lfloor \frac{j}{4} \right\rfloor\right)\left(1 + \left\lfloor \frac{j+2}{4} \right\rfloor\right).$$

Suppose that $j = 4j_0 + r$ with $0 \leq r \leq 3$. Let $\mathcal{P}_j$ denote the set of partitions of j with parts $1, 2$ or 4. We consider the following four cases according to the values of r.

Case 1: If $r = 0$, then $j = 4j_0$. For any $\lambda = (4, \ldots, 4, 2, \ldots, 2, 1, \ldots, 1) \in \mathcal{P}_j$, we see that the number of 1's must be even. We take a map τ_1: $\mathcal{P}_j \to \mathcal{P}_{j+1}$ by $\tau_1(\lambda) = \lambda_1$, where λ_1 is given by adding one more 1 part into λ. Thus $\lambda_1 \in \mathcal{P}_{j+1}$ with odd number of 1's. It is straightforward to see that τ_1 is a bijection between $\mathcal{P}_j$ and $\mathcal{P}_{j+1}$. By the assumption, we have

$$p_4(j+1) = p_4(j) = (j_0 + 1)^2.$$

Case 2: If $r = 1$, then $j = 4j_0 + 1$. For any $\lambda = (4, \ldots, 4, 2, \ldots, 2, 1, \ldots, 1) \in \mathcal{P}_j$, it has odd number of 1's. We define an injection τ_2: $\mathcal{P}_j \to \mathcal{P}_{j+1}$ by $\tau_2(\lambda) = \lambda_2$, where $\lambda_2 \in \mathcal{P}_{j+1}$ is given by adding one to the first 1 part of λ. Then λ_2 has even number of 1's (which may be zero) and at least one part is 2.

Denote the set $\mathcal{P}_{j+1} \setminus \tau_2(\mathcal{P}_j) = \mathcal{P}'_{j+1}$. Then any partition $\mu \in \mathcal{P}'_{j+1}$ is a partition belonging to $\mathcal{P}_{j+1}$ with no part being 2. Note that in this case $j + 1 = 4j_0 + 2$, then μ can be partitioned as follows

$$\mu = j_0 \times 4 + 2 \times 1 = (j_0 - 1) \times 4 + 6 \times 1 = (j_0 - 2) \times 4 + 8 \times 1 = \cdots = (4j_0 + 2) \times 1.$$

It implies that the number of such partitions in $\mathcal{P}'_{j+1}$ with no 2's is $j_0 + 1$, i.e. $|\mathcal{P}'_{j+1}| = j_0 + 1$. By noting that $\mathcal{P}_{j+1} = \tau(\mathcal{P}_j) \bigcup \mathcal{P}'_{j+1}$ and $\tau(\mathcal{P}_j) \bigcap \mathcal{P}'_{j+1} = \emptyset$, we are led to

$$\begin{aligned} p_4(j+1) &= p_4(j) + (j_0 + 1) \\ &= (j_0 + 1)(j_0 + 1) + (j_0 + 1) = (j_0 + 1)(j_0 + 2). \end{aligned}$$

Case 3: If $r = 2$, then $j = 4j_0 + 2$. By the bijection τ_1 as given in Case 1 and under the assumption, we have in this case

$$p_4(j+1) = p_4(j) = (j_0 + 1)(j_0 + 2).$$

Case 4: If $r = 3$, then $j = 4j_0 + 3$. Following similar discussions under the injection τ_2 as given in Case 2, we have

$$\begin{aligned} p_4(j+1) &= p_4(j) + (j_0 + 2) \\ &= (j_0 + 1)(j_0 + 2) + (j_0 + 2) = (j_0 + 2)^2. \end{aligned}$$

For the above four cases, we see that all the expressions of $p_4(j+1)$ can be written as

$$p_4(j+1)=\left(1+\left\lfloor\frac{j+1}{4}\right\rfloor\right)\left(1+\left\lfloor\frac{(j+1)+2}{4}\right\rfloor\right),$$

and thereby the proof is completed by induction. ∎

3. Proof of Ballantine and Merca's conjecture

In this section, we shall prove Conjecture 1.4 and extend it to more general cases. The following result plays an important role in our proof. Moreover, note that n, n_0, and k are assumed to be nonnegative integers.

Theorem 3.1. *For $n\geq 0$, $k\geq 2$, the coefficients of q^n in the series*

$$\frac{1}{(1-q)(1-q^2)(1-q^4)}\sum_{j=0}^{\infty}(-1)^j q^{\left(3j^2+6kj+j\right)/2}\left(1-q^{2j+2k+1}\right) \tag{3.1}$$

are positive.

Proof. By the generating function (2.4) for $p_4(n)$, we can write the coefficients of q^n in (3.1) as follows

$$C(n)=\sum_{j=0}^{\infty}(-1)^j\left(p_4\Big(n-\frac{3j^2+6kj+j}{2}\Big)-p_4\Big(n-\frac{3j^2+6kj+5j+4k+2}{2}\Big)\right),$$

in which there are finite number of nonzero summands since $p_4(n)=0$ if $n<0$. Then by considering the parities of j, it turns to be

$$\begin{aligned}C(n)=\sum_{j=0}^{\infty}\Big(&p_4\big(n-6j^2-6kj-j\big)-p_4\big(n-6j^2-6kj-5j-2k-1\big)\\&-p_4\big(n-6j^2-6kj-7j-3k-2\big)+p_4\big(n-6j^2-6kj-11j-5k-5\big)\Big).\end{aligned} \tag{3.2}$$

We consider the partial sum of the first n_0 terms in $C(n)$ and denote it by

$$\begin{aligned}C_{n_0}(n)=\sum_{j=0}^{n_0-1}\Big(&p_4\big(n-6j^2-6kj-j\big)-p_4\big(n-6j^2-6kj-5j-2k-1\big)\\&-p_4\big(n-6j^2-6kj-7j-3k-2\big)+p_4\big(n-6j^2-6kj-11j-5k-5\big)\Big).\end{aligned} \tag{3.3}$$

Note that the set of nonnegative integers can be decomposed as follows

$$\bigcup_{n_0\geq 0}\big\{x\in\mathbb{N}\colon 6{n_0}^2+6kn_0+n_0\leq x<6(n_0+1)^2+6k(n_0+1)+n_0+1\big\}.$$

Thus to show that $C(n)>0$ for $n\geq 0$, it suffices to verify that for any given $n_0\geq 0$, it holds for n belonging to the interval

$$6n_0^2+6kn_0+n_0\leq n<6(n_0+1)^2+6k(n_0+1)+n_0+1.$$

Denote

$$\begin{aligned} f_1(n_0,k) &= 6n_0^2+6kn_0+n_0,\\ f_2(n_0,k) &= 6n_0^2+6kn_0+5n_0+2k+1,\\ f_3(n_0,k) &= 6n_0^2+6kn_0+7n_0+3k+2,\\ f_4(n_0,k) &= 6n_0^2+6kn_0+11n_0+5k+5, \end{aligned}$$

then the interval can be further decomposed as follows

$$\left[6n_0^2+6kn_0+n_0, 6(n_0+1)^2+6k(n_0+1)+n_0\right] = \bigcup_{i=1}^{4} \left[f_i(n_0,k), f_{i+1}(n_0,k)\right),$$

where $f_5(n_0,k) = f_1(n_0+1,k)$.

By using the explicit formula (2.5) for $p_4(n)$, when n belongs to each of the above intervals $[f_i(n_0,k), f_{i+1}(n_0,k))$, we need to explore 64 subcases according to the residues of n, n_0, and k modulo 4, respectively. By calculating all these subcases, we find that only when $k=1$, (3.1) may have negative coefficients. In the appendix, we state the corresponding results for $C(n)$ when $k \equiv 1 \pmod 4$. For other cases, please see `https://github.com/IgnisDYChen/Appendix` for details. Note that in the labeling of the subcases, the first level corresponds to the interval under consideration, the second level indicates the remaining class of k modulo 4, and the third level represents the sixteen cases of n and n_0 modulo 4.

Then to show $C(n)$ is positive for $n \geq 0$, it is sufficient to consider the minimum value of $C(n)$. Since all the subcases can be analyzed similarly, here we only state some of them in detail.

Case 1: $f_1(n_0,k) \leq n < f_2(n_0,k)$.

In this case, from (3.2) and (3.3), it is easy to see that

$$C(n) = C_{n_0}(n) + p_4(n - f_1(n_0,k)). \tag{3.4}$$

According to the residues of n, n_0 and k modulo 4, we would further decompose this case into 64 subcases, for which we state the following three cases.

Subcase 1.1.1: $(n, n_0, k) \equiv (0,0,0) \pmod 4$.

Denote that $n = 4m$, $n_0 = 4m_0$ and $k = 4k_0$, where m, m_0 and k_0 are suppose to be nonnegative integers throughout this proof. Then we have $f_1(4m_0, 4k_0) \leq 4m < f_2(4m_0, 4k_0)$, which leads to that

$$24m_0^2+24k_0m_0+m_0 \leq m < 24m_0^2+24k_0m_0+5m_0+2k_0+\frac{1}{4}.$$

By (3.4) and (2.5), we have

$$\begin{aligned} C(n) &= C_{4m_0}(4m) + p_4(4m - f_1(4m_0,4k_0))\\ &= m^2 - (48k_0m_0+48m_0^2+6m_0-2)m + 576k_0^2m_0^2+48k_0^2m_0+1152k_0m_0^3\\ &\quad + 192k_0m_0^2 - 50k_0m_0 + 576m_0^4+144m_0^3-47m_0^2-8m_0+1. \end{aligned}$$

Note that the above result is a quadratic function of m. If $m_0 = 0$, then $C(n) = m^2+2m+1$, which directly implies that $C(n) > 0$. To show that $C(n) > 0$, we only need to analyze the values of the function at the endpoints of the interval as well as

on the symmetry axis. For $m_0 \geq 1$, $k_0 \geq 1$, we can see that the minimum value of this quadratic function satisfies that

$$C(24k_0m_0 + 24m_0^2 + 3m_0 - 1) = (48k_0 - 8)m_0^2 + (48k_0^2 - 2k_0 - 2)m_0 > 0.$$

Thus in this case $C(n) > 0$ when $n \geq 0$ and $k \geq 4$ with $k \equiv 0 \pmod 4$.

Subcase 1.2.1: $(n, n_0, k) \equiv (0, 0, 1) \pmod 4$.

Suppose that $n = 4m$, $n_0 = 4m_0$ and $k = 4k_0 + 1$, we have $f_1(4m_0, 4k_0 + 1) \leq 4m < f_2(4m_0, 4k_0 + 1)$, it follows that

$$24m_0^2 + 24k_0m_0 + 7m_0 \leq m < 24m_0^2 + 24k_0m_0 + 11m_0 + 2k_0 + \frac{3}{4}.$$

By (3.4), we have

$$\begin{aligned} C(n) &= C_{4m_0}(4m) + p_4(4m - f_1(4m_0, 4k_0 + 1)) \\ &= m^2 + (-48k_0m_0 - 48m_0^2 - 18m_0 + 2)m + 576k_0^2m_0^2 + 48k_0^2m_0 \\ &\quad + 1152k_0m_0^3 + 480k_0m_0^2 - 14k_0m_0 + 576m_0^4 + 432m_0^3 + 49m_0^2 - 13m_0 + 1. \end{aligned}$$

It is direct to see that $C(n) > 0$ when $m_0 = 0$. Then we just need to analyze the values of the above function at the endpoints of the interval as well as on the symmetry axis. Then in this case for $m_0 \geq 1$ and $k_0 \geq 0$,

$$C(n) \geq C(24m_0^2 + 24k_0m_0 + 9m_0 - 1) = (48k_0 + 16)m_0^2 + (48k_0^2 + 34k_0 + 5)m_0 > 0.$$

Subcase 1.2.6: $(n, n_0, k) \equiv (1, 1, 1) \pmod 4$.

Let $n = 4m+1$, $n_0 = 4m_0+1$ and $k = 4k_0+1$. Then we have $f_1(4m_0+1, 4k_0+1) \leq 4m + 1 < f_2(4m_0 + 1, 4k_0 + 1)$, which implies that

$$24m_0^2 + 24k_0m_0 + 6k_0 + 19m_0 + 3 \leq m < 24m_0^2 + 24k_0m_0 + 8k_0 + 23m_0 + 3.$$

Then by (3.4), we have

$$\begin{aligned} C(n) &= C_{4m_0+1}(4m + 1) + p_4(4m + 1 - f_1(4m_0 + 1, 4k_0 + 1)) \\ &= m^2 + (-48m_0^2 - 48k_0m_0 - 12k_0 - 42m_0 - 6)m + 576k_0^2m_0^2 + 336k_0^2m_0 \\ &\quad + 48k_0^2 + 1152k_0m_0^3 + 1344k_0m_0^2 + 430k_0m_0 + 41k_0 + 576m_0^4 + 1008m_0^3 \\ &\quad + 577m_0^2 + 118m_0 + 7. \end{aligned}$$

Denote the above function as a quadratic function $F_1(m)$ of m, then the value of $F_1(m)$ at the symmetry axis is as follows:

$$\begin{aligned} &F_1(24m_0^2 + 24k_0m_0 + 6k_0 + 21m_0 + 3) \\ &\quad = (48k_0 - 8)m_0^2 + (48k_0^2 + 34k_0 - 8)m_0 + 12k_0^2 + 5k_0 - 2. \end{aligned}$$

The value of $F_1(m)$ at the left endpoint of the interval is as follows:

$$\begin{aligned} &F_1\left(\lceil (f_1(4m_0 + 1, 4k_0 + 1) - 1)/4 \rceil\right) \\ &\quad = f(24m_0^2 + 24k_0m_0 + 6k_0 + 19m_0 + 3) \\ &\quad = (48k_0 - 4)m_0^2 + (48k_0^2 + 34k_0 - 8)m_0 + 12k_0^2 + 5k_0 - 2, \end{aligned}$$

where $\lceil a \rceil$ represents the ceiling function.

The value of $F_1(m)$ at the right endponit of the interval is as follows:

$$\begin{aligned}&F_1\left(\lfloor (f_2(4m_0+1,4k_0+1)-2)/4\rfloor\right)\\&\quad=f(24m_0^2+24k_0m_0+8k_0+23m_0+2)\\&\quad=(48k_0-4)m_0^2+(48k_0^2+42k_0-16)m_0+16k_0^2-3k_0-14.\end{aligned}$$

The above three values are greater than 0 except for $k_0=0$, i.e., $k=1$. Thus for any $k_0\geq 1$, that is $k\geq 5$ and $k\equiv 1 \pmod 4$, $C(n)>0$.

After inspecting all the 64 subcases, we see that when $m\geq 0$, $k\geq 2$, $C(n)>0$ for $n\in[f_1(n_0,k),f_2(n_0,k))$.

Case 2: $f_2(n_0,k)\leq n<f_3(n_0,k)$.

In this case, from (3.2) and (3.3), it follows that

$$C(n)=C_{n_0}(n)+p_4(n-f_1(n_0,k))-p_4(n-f_2(n_0,k)-1). \tag{3.5}$$

Subcase 2.2.2: $(n,n_0,k)\equiv(0,1,1) \pmod 4$.

Let $n=4m$, $n_0=4m_0+1$ and $k=4k_0+1$, by (3.5) we have

$$\begin{aligned}C(n)=C(4m)=(4k_0+4m_0+3)m-48k_0^2m_0-16k_0^2-144k_0m_0^2-130k_0m_0\\-22k_0-96m_0^3-144m_0^2-64m_0-7=:F_2(m).\end{aligned}$$

We see that $F_2(m)$ is a linear function of m, and the value of $F_2(m)$ at the left endpoint of the interval is

$$\begin{aligned}&F_2\left(\lceil f_2(4m_0+1,4k_0+1)/4\rceil\right)=F_2(24m_0^2+24k_0m_0+8k_0+23m_0+5)\\&\quad=(48k_0+20)m_0^2+(48k_0^2+66k_0+25)m_0+16k_0^2+22k_0+8.\end{aligned}$$

Thus for all $k_0\geq 0$ ($k\geq 1$ and $k\equiv 1 \pmod 4$), $C(n)>0$.

Subcase 2.2.5: $(n,n_0,k)\equiv(1,0,1) \pmod 4$.

Let $n=4m+1$, $n_0=4m_0$ and $k=4k_0+1$, by (3.5) we have

$$\begin{aligned}C(n)=C(4m+1)=(4k_0+4m_0+1)m-48k_0^2m_0-4k_0^2-144k_0m_0^2-58k_0m_0+2k_0\\-96m_0^3-72m_0^2-9m_0+1=:F_3(m).\end{aligned}$$

Note that $F_3(m)$ is a linear function of m, and the value of $F_3(m)$ at the left endpoint of the interval is as follows

$$\begin{aligned}&F_3\left(\lceil (f_2(4m_0,4k_0+1)-1)/4\rceil\right)=F_3(24m_0^2+24k_0m_0+11m_0+2k_0+1)\\&\quad=(48k_0-4)m_0^2+(48k_0^2+18k_0+6)m_0+4k_0^2+8k_0+2.\end{aligned}$$

Thus for all $k_0\geq 1$, i.e., $k\geq 5$ and $k\equiv 1 \pmod 4$, $C(n)>0$.

Subcase 2.3.2: $(n,n_0,k)\equiv(0,1,2) \pmod 4$.

Let $n=4m$, $n_0=4m_0+1$ and $k=4k_0+2$, by (3.5) we have

$$\begin{aligned}&C(n)=C(4m)\\&\quad=(4k_0+4m_0+3)n-48k_0^2m_0-16k_0^2-144k_0m_0^2-142k_0m_0-26k_0-96m_0^3\\&\qquad-168m_0^2-86m_0-11=:F_4(m).\end{aligned}$$

Since $F_4(m)$ is a linear function, we only need to compute the left endpoint of the interval:

$$F_4\left(\lceil f_2(4m_0+1,4k_0+2)/4\rceil\right) = F_4(24m_0^2+24k_0m_0+29m_0+8k_0+7)$$
$$= (48k_0^2+78k_0+29)m_0+16k_0^2+(48k_0+20)m_0^2+26k_0+10,$$

It is obvious that for all $k_0 \geq 0$ ($k \geq 2$ and $k \equiv 2 \pmod 4$), $C(n) > 0$.

After inspecting all the 64 subcases, we show that when $m \geq 0$, $k \geq 2$, $C(n) > 0$ for $n \in [f_2(n_0,k), f_3(n_0,k))$.

Case 3: $f_3(n_0,k) \leq n < f_4(n_0,k)$.

In this case, from (3.2) and (3.3), we can see that

(3.6)
$$C(n) = C_{n_0}(n) + p_4(n - f_1(n_0,k)) - p_4(n - f_2(n_0,k) - 1) - p_4(n - f_3(n_0,k) - 1).$$

Subcase 3.2.12: $(n, n_0, k) \equiv (2,3,1) \pmod 4$.

Let $n = 4m+2$, $n_0 = 4m_0+3$ and $k = 4k_0+1$, by (3.6) we have

$$\begin{aligned} C(n) = C(4m+2) = &-m^2 + (48k_0m_0+46k_0+48m_0^2+102m_0+51)m \\ &- 576k_0^2m_0^2 - 1056k_0^2m_0 - 481k_0^2 - 1152k_0m_0^3 - 3504k_0m_0^2 \\ &- 3436k_0m_0 - 1087k_0 - 576m_0^4 - 2448m_0^3 - 3793m_0^2 \\ &- 2532m_0 - 613 =: F_5(m). \end{aligned}$$

It is not difficult to find that the minimum value of $F_5(m)$ is taken at the left endpoint of the interval which is

$$\begin{aligned} &F_5\left(\lceil (f_3(4m_0+3,4k_0+1)-2)/4\rceil\right) \\ &= F_5(24m_0^2+24k_0m_0+21k_0+49m_0+24) \\ &= 1152m_0^4+(2304k_0+4704)m_0^3+(1152k_0^2+6768k_0+7134)m_0^2 \\ &\quad +(2064k_0^2+6546k_0+4767)m_0+926k_0^2+2096k_0+1187. \end{aligned}$$

It implies that for all $k_0 \geq 0$ ($k \geq 1$ and $k \equiv 1 \pmod 4$), $C(n) > 0$.

After inspecting all the 64 subcases, we see that when $m \geq 0$, $k \geq 2$, $C(n) > 0$ for $n \in [f_3(n_0,k), f_4(n_0,k))$.

Case 4: $f_4(n_0,k) \leq n < f_1(n_0+1,k)$.

In this case, from (3.2) and (3.3), we have

(3.7)
$$C(n) = C_{n_0+1}(n).$$

Subcase 4.2.16: $(n, n_0, k) \equiv (3,3,1) \pmod 4$.

Let $n = 4m+3$, $n_0 = 4m_0+3$ and $k = 4k_0+1$, by (3.7) we have

$$\begin{aligned} C(n) = C(4m+3) = &(-4m_0-4)m + 48k_0^2m_0 + 48k_0^2 + 144k_0m_0^2 + 298k_0m_0 \\ &+ 154k_0 + 96m_0^3 + 312m_0^2 + 328m_0 + 112 =: F_6(m), \end{aligned}$$

Since $F_6(m)$ is a decreasing linear function of m, it is sufficient to see that the value of $F_6(m)$ at the right endpoint of the interval

$$F_6\left(\lfloor (f_1(4m_0+5,4k_0+1)-4)/4\rfloor\right)$$
$$= F_5(24k_0m_0+24k_0+24m_0^2+55m_0+29)$$

$$= 48k_0^2m_0 + 48k_0^2 + 48k_0m_0^2 + 106k_0m_0 + 58k_0 - 4m_0^2 - 12m_0 - 8.$$

Thus for all $k_0 \geq 1$ ($k \geq 5$ and $k \equiv 1 \pmod 4$), $C(n) > 0$.

Subcase 4.4.10: $(n, n_0, k) \equiv (2, 1, 3) \pmod 4$.

Let $n = 4m + 2$, $n_0 = 4m_0 + 1$ and $k = 4k_0 + 3$, by (3.7) we have

$$\begin{aligned} C(n) = C(4m_0 + 2) =& (-4m_0 - 2)m + 48k_0^2m_0 + 24k_0^2 + 144k_0m_0^2 + 226k_0m_0 \\ &+ 77k_0 + 96m_0^3 + 264m_0^2 + 220m_0 + 56 =: F_7(m), \end{aligned}$$

Since $F_7(m)$ is a linear function of m, and the value of $F_7(m)$ at the right endpoint of the interval is

$$\begin{aligned} &F_7\left(\lfloor (f_1(4m_0 + 2, 4k_0 + 3) - 3)/4 \rfloor\right) \\ &\quad = F_7(24k_0m_0 + 12k_0 + 24m_0^2 + 31m_0 + 9) \\ &\quad = 48k_0^2m_0 + 24k_0^2 + 48k_0m_0^2 + 130k_0m_0 + 53k_0 + 92m_0^2 + 126m_0 + 40. \end{aligned}$$

It is obviously that for all $k_0 \geq 0$ ($k \geq 3$ and $k \equiv 3 \pmod 4$), $C(n) > 0$.

From the appendix, after inspecting all the 64 subcases, we show that when $m \geq 0$, $k \geq 2$, $C(n) > 0$ for $n \in [f_4(n_0, k), f_1(n_0 + 1, k))$.

Above all, we complete the proof. ∎

Now we are ready to extend Ballantine and Merca's Conjecture 1.4 to more general cases with respected to ℓ.

Theorem 3.2. *For $n \geq 0$, $k \geq 1$, $\ell = 3$ or $\ell \geq 5$, the coefficients of q^n in*

$$(-1)^k \Big(\left(q^\ell; q^\ell\right)_\infty - \frac{\left(q^\ell; q^\ell\right)_\infty}{(q; q)_\infty} \sum_{j=-k+1}^{k} (-1)^j q^{j(3j-1)/2} \Big) \tag{3.8}$$

are all zero for $0 \leq n < k(3k+1)/2$, and are positive for $n \geq k(3k+1)/2$.

Proof. Firstly, by using Euler's pentagonal number theorem (1.1), we can rewrite (3.8) as follows

$$\begin{aligned} &(-1)^k \Big(\left(q^\ell; q^\ell\right)_\infty - \frac{\left(q^\ell; q^\ell\right)_\infty}{(q; q)_\infty} \sum_{j=-k+1}^{k} (-1)^j q^{j(3j-1)/2} \Big) \\ &= (-1)^k \frac{\left(q^\ell; q^\ell\right)_\infty}{(q; q)_\infty} \Big(\sum_{j=-\infty}^{\infty} - \sum_{j=-k+1}^{k} \Big) (-1)^j q^{j(3j-1)/2} \\ &= (-1)^k \frac{\left(q^\ell; q^\ell\right)_\infty}{(q; q)_\infty} \Big(\sum_{j=-\infty}^{-k} + \sum_{j=k+1}^{\infty} \Big) (-1)^j q^{j(3j-1)/2} \\ &= (-1)^k \frac{\left(q^\ell; q^\ell\right)_\infty}{(q; q)_\infty} \sum_{j=k}^{\infty} (-1)^j q^{3j(j+1)/2} (q^{-j} - q^{j+1}) \qquad (3.9) \\ &= (-1)^k \frac{\left(q^\ell; q^\ell\right)_\infty}{(q^3; q^3)_\infty} \frac{\sum_{j=k}^{\infty} (-1)^j q^{3j(j+1)/2} (q^{-j} - q^{j+1})}{(q, q^2; q^3)_\infty}. \qquad (3.10) \end{aligned}$$

When $k=1$ and $\ell \geq 3$, by substituting $R=3S$ and $S=1$ into (1.7) and applying Ballantine and Feigon's result, it directly leads to the coefficients of q^n in (3.8) satisfy the required conditions.

For $k \geq 2$, (3.8) can be further rewritten as

$$\begin{aligned}
&(-1)^k\Big(\left(q^\ell;q^\ell\right)_\infty - \frac{\left(q^\ell;q^\ell\right)_\infty}{(q;q)_\infty} \sum_{j=-k+1}^{k} (-1)^j q^{j(3j-1)/2}\Big) \\
&= q^{\frac{3k^2+k}{2}} \frac{\left(q^\ell;q^\ell\right)_\infty}{(q;q)_\infty} \sum_{j=0}^{\infty} (-1)^j q^{\left(3j^2+6kj+j\right)/2} \left(1-q^{2j+2k+1}\right) \\
&= q^{\frac{3k^2+k}{2}} \frac{\left(q^\ell;q^\ell\right)_\infty}{\left(1-q^3\right)\left(q^5;q\right)_\infty} \frac{\sum_{j=0}^{\infty} (-1)^j q^{\left(3j^2+6jk+j\right)/2} \left(1-q^{2j+2k+1}\right)}{(1-q)(1-q^2)(1-q^4)}.
\end{aligned}$$

Obviously, we can see that the coefficients of q^n are all zero if $n < k(3k+1)/2$. Moreover, when $\ell = 3$ or $\ell \geq 5$, the coefficients in

$$\frac{\left(q^\ell;q^\ell\right)_\infty}{\left(1-q^3\right)\left(q^5;q\right)_\infty}$$

are nonnegative. Combining it with Theorem 3.1, the proof is complete. ∎

Clearly, when $\ell = 6$, Ballantine and Merca's Conjecture 1.4 for 6-regular partitions is confirmed. We also remark that from Yao's result [16, Theorem 1.3], following the similar procedures as given above, it can be shown that Theorem 3.2 still holds for $\ell = 4$. Therefore, we can conclude that when $\ell \geq 3$, the coefficients of q^n in (3.8) are nonnegative and satisfy the conditions as given in Theorem 3.2.

From the above theorem, we can obtain the following positivity property on partitions with parts not congruent to $0, \pm 3 \pmod{12}$, which was conjectured by Merca in [12, Conjecture 4.1].

Corollary 3.3. *For $k \geq 1$, the coefficients of q^n in the series*

$$(-1)^{k-1}\left(\frac{1}{(q;q)_\infty} \sum_{j=-k+1}^{k} (-1)^j q^{j(3j-1)/2} - 1\right)\left(q^3,q^9;q^{12}\right)_\infty$$

are all zero for $0 \leq n < k(3k+1)/2$, and are positive for $n \geq k(3k+1)/2$.

Proof. From (3.9) and by substituting $\ell = 3$, we obtain

$$\begin{aligned}
&(-1)^{k-1}\left(\frac{1}{(q;q)_\infty} \sum_{j=-k+1}^{k} (-1)^j q^{j(3j-1)/2} - 1\right)\left(q^3,q^9;q^{12}\right)_\infty \\
&= (-1)^k \left(\left(q^3;q^3\right)_\infty - \frac{\left(q^3;q^3\right)_\infty}{(q;q)_\infty} \sum_{j=-k+1}^{k} (-1)^j q^{j(3j-1)/2} \right) \frac{\left(q^3,q^9;q^{12}\right)_\infty}{\left(q^3;q^3\right)_\infty} \\
&= \frac{1}{\left(q^6,q^{12};q^{12}\right)_\infty} \frac{\left(q^3;q^3\right)_\infty}{(q;q)_\infty} (-1)^k \sum_{j\geq k} (-1)^j q^{j(3j+1)/2}(1-q^{2j+1}). \qquad (3.11)
\end{aligned}$$

From (3.11), it can be seen that the coefficients of q^n are all zero for $0 \leq n < k(3k+1)/2$. Substituting $\ell = 3$ into Theorem 3.2, it completes the proof. ∎

Finally, we show that Theorem 3.2 also implies to that when $R = 3S$, Conjecture 1.3 holds.

Corollary 3.4. *Conjecture 1.3 holds when $R = 3S$ and $S \geq 1$.*

Proof. Substituting $\ell = 3$ into (3.10) in the proof of Theorem 3.2, it directly leads to that

$$(-1)^k \frac{\sum_{j=k}^{\infty}(-1)^j q^{3j(j+1)/2}\left(q^{-j} - q^{j+1}\right)}{(q, q^2; q^3)_\infty}$$

has nonnegative coefficients of q^n. By replacing q with q^S in the above expression, the proof is complete. ∎

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Case	(n, n_0, k)	$C(n)$
1.2.1	$(0,0,1)$	$m^2+(-48k_0m_0-48m_0^2-18m_0+2)m+576k_0^2m_0^2+48k_0^2m_0+1152k_0m_0^3+480k_0m_0^2-14k_0m_0+576m_0^4+432m_0^3+49m_0^2-13m_0+1$
1.2.2	$(0,1,1)$	$m^2+(-48k_0m_0-12k_0-48m_0^2-42m_0-5)m+576k_0^2m_0^2+336k_0^2m_0+48k_0^2+1152k_0m_0^3+1344k_0m_0^2+430k_0m_0+42k_0+576m_0^4+1008m_0^3+577m_0^2+120m_0+9$
1.2.3	$(0,2,1)$	$m^2+(-48k_0m_0-24k_0-48m_0^2-66m_0-19)m+576k_0^2m_0^2+624k_0^2m_0+168k_0^2+1152k_0m_0^3+2208k_0m_0^2+1330k_0m_0+257k_0+576m_0^4+1584m_0^3+1561m_0^2+648m_0+96$
1.2.4	$(0,3,1)$	$m^2+(-48k_0m_0-36k_0-48m_0^2-90m_0-38)m+576k_0^2m_0^2+912k_0^2m_0+360k_0^2+1152k_0m_0^3+3072k_0m_0^2+2638k_0m_0+737k_0+576m_0^4+2160m_0^3+2953m_0^2+1741m_0+375$
1.2.5	$(1,0,1)$	$m^2+(-48k_0m_0-48m_0^2-18m_0+2)m+576k_0^2m_0^2+48k_0^2m_0+1152k_0m_0^3+480k_0m_0^2-38k_0m_0+576m_0^4+432m_0^3+25m_0^2-20m_0+1$
1.2.6	$(1,1,1)$	$m^2+(-48k_0m_0-12k_0-48m_0^2-42m_0-6)m+576k_0^2m_0^2+336k_0^2m_0+48k_0^2+1152k_0m_0^3+1344k_0m_0^2+430k_0m_0+41k_0+576m_0^4+1008m_0^3+577m_0^2+118m_0+7$
1.2.7	$(1,2,1)$	$m^2+(-48k_0m_0-24k_0-48m_0^2-66m_0-19)m+576k_0^2m_0^2+624k_0^2m_0+168k_0^2+1152k_0m_0^3+2208k_0m_0^2+1306k_0m_0+245k_0+576m_0^4+1584m_0^3+1537m_0^2+617m_0+87$
1.2.8	$(1,3,1)$	$m^2+(-48k_0m_0-36k_0-48m_0^2-90m_0-39)m+576k_0^2m_0^2+912k_0^2m_0+360k_0^2+1152k_0m_0^3+3072k_0m_0^2+2638k_0m_0+736k_0+576m_0^4+2160m_0^3+2953m_0^2+1739m_0+372$
1.2.9	$(2,0,1)$	$m^2+(-48k_0m_0-48m_0^2-18m_0+3)m+576k_0^2m_0^2+48k_0^2m_0+1152k_0m_0^3+480k_0m_0^2-38k_0m_0+576m_0^4+432m_0^3+25m_0^2-22m_0+2$
1.2.10	$(2,1,1)$	$m^2+(-48k_0m_0-12k_0-48m_0^2-42m_0-4)m+576k_0^2m_0^2+336k_0^2m_0+48k_0^2+1152k_0m_0^3+1344k_0m_0^2+406k_0m_0+36k_0+576m_0^4+1008m_0^3+553m_0^2+99m_0+7$
1.2.11	$(2,2,1)$	$m^2+(-48k_0m_0-24k_0-48m_0^2-66m_0-18)m+576k_0^2m_0^2+624k_0^2m_0+168k_0^2+1152k_0m_0^3+2208k_0m_0^2+1306k_0m_0+245k_0+576m_0^4+1584m_0^3+1537m_0^2+615m_0+88$
1.2.12	$(2,3,1)$	$m^2+(-48k_0m_0-36k_0-48m_0^2-90m_0-37)m+576k_0^2m_0^2+912k_0^2m_0+360k_0^2+1152k_0m_0^3+3072k_0m_0^2+2614k_0m_0+719k_0+576m_0^4+2160m_0^3+2929m_0^2+1696m_0+357$
1.2.13	$(3,0,1)$	$m^2+(-48k_0m_0-48m_0^2-18m_0+3)m+576k_0^2m_0^2+48k_0^2m_0+1152k_0m_0^3+480k_0m_0^2-62k_0m_0+576m_0^4+432m_0^3+m_0^2-29m_0+2$
1.2.14	$(3,1,1)$	$m^2+(-48k_0m_0-12k_0-48m_0^2-42m_0-5)m+576k_0^2m_0^2+336k_0^2m_0+48k_0^2+1152k_0m_0^3+1344k_0m_0^2+406k_0m_0+35k_0+576m_0^4+1008m_0^3+553m_0^2+97m_0+4$
1.2.15	$(3,2,1)$	$m^2+(-48k_0m_0-24k_0-48m_0^2-66m_0-18)m+576k_0^2m_0^2+624k_0^2m_0+168k_0^2+1152k_0m_0^3+2208k_0m_0^2+1282k_0m_0+233k_0+576m_0^4+1584m_0^3+1513m_0^2+584m_0+78$
1.2.16	$(3,3,1)$	$m^2+(-48k_0m_0-36k_0-48m_0^2-90m_0-38)m+576k_0^2m_0^2+912k_0^2m_0+360k_0^2+1152k_0m_0^3+3072k_0m_0^2+2614k_0m_0+718k_0+576m_0^4+2160m_0^3+2929m_0^2+1694m_0+353$
2.2.1	$(0,0,1)$	$(4k_0+4m_0+2)m-48k_0^2m_0-4k_0^2-144k_0m_0^2-58k_0m_0-96m_0^3-72m_0^2-13m_0+1$
2.2.2	$(0,1,1)$	$(4k_0+4m_0+3)m-48k_0^2m_0-16k_0^2-144k_0m_0^2-130k_0m_0-22k_0-96m_0^3-144m_0^2-64m_0-7$
2.2.3	$(0,2,1)$	$(4k_0+4m_0+4)m-48k_0^2m_0-28k_0^2-144k_0m_0^2-202k_0m_0-65k_0-96m_0^3-216m_0^2-157m_0-36$
2.2.4	$(0,3,1)$	$(4k_0+4m_0+5)m-48k_0^2m_0-40k_0^2-144k_0m_0^2-274k_0m_0-123k_0-96m_0^3-288m_0^2-280m_0-87$
2.2.5	$(1,0,1)$	$(4k_0+4m_0+1)m-48k_0^2m_0-4k_0^2-144k_0m_0^2-58k_0m_0+2k_0-96m_0^3-72m_0^2-9m_0+1$
2.2.6	$(1,1,1)$	$(4k_0+4m_0+2)m-48k_0^2m_0-16k_0^2-144k_0m_0^2-130k_0m_0-23k_0-96m_0^3-144m_0^2-66m_0-9$
2.2.7	$(1,2,1)$	$(4k_0+4m_0+3)m-48k_0^2m_0-28k_0^2-144k_0m_0^2-202k_0m_0-63k_0-96m_0^3-216m_0^2-153m_0-34$
2.2.8	$(1,3,1)$	$(4k_0+4m_0+4)m-48k_0^2m_0-40k_0^2-144k_0m_0^2-274k_0m_0-124k_0-96m_0^3-288m_0^2-282m_0-90$
2.2.9	$(2,0,1)$	$(4k_0+4m_0+2)m-48k_0^2m_0-4k_0^2-144k_0m_0^2-58k_0m_0+2k_0-96m_0^3-72m_0^2-11m_0+2$
2.2.10	$(2,1,1)$	$(4k_0+4m_0+3)m-48k_0^2m_0-16k_0^2-144k_0m_0^2-130k_0m_0-20k_0-96m_0^3-144m_0^2-62m_0-5$
2.2.11	$(2,2,1)$	$(4k_0+4m_0+4)m-48k_0^2m_0-28k_0^2-144k_0m_0^2-202k_0m_0-63k_0-96m_0^3-216m_0^2-155m_0-33$
2.2.12	$(2,3,1)$	$(4k_0+4m_0+5)m-48k_0^2m_0-40k_0^2-144k_0m_0^2-274k_0m_0-121k_0-96m_0^3-288m_0^2-278m_0-84$
2.2.13	$(3,0,1)$	$(4k_0+4m_0+1)m-48k_0^2m_0-4k_0^2-144k_0m_0^2-58k_0m_0+4k_0-96m_0^3-72m_0^2-7m_0+1$
2.2.14	$(3,1,1)$	$(4k_0+4m_0+2)m-48k_0^2m_0-16k_0^2-144k_0m_0^2-130k_0m_0-21k_0-96m_0^3-144m_0^2-64m_0-8$

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Table 1 – continued from previous page

Case	(n, n_0, k)	$C(n)$
2.2.15	$(3,2,1)$	$(4k_0+4m_0+3)m-48k_0^2m_0-28k_0^2-144k_0m_0^2-202k_0m_0-61k_0-96m_0^3-216m_0^2-151m_0-32$
2.2.16	$(3,3,1)$	$(4k_0+4m_0+4)m-48k_0^2m_0-40k_0^2-144k_0m_0^2-274k_0m_0-122k_0-96m_0^3-288m_0^2-280m_0-88$
3.2.1	$(0,0,1)$	$-m^2+(48k_0m_0+10k_0+48m_0^2+30m_0+3)m-576k_0^2m_0^2-192k_0^2m_0-13k_0^2-1152k_0m_0^3-912k_0m_0^2-160k_0m_0-3k_0-576m_0^4-720m_0^3-265m_0^2-26m_0+1$
3.2.2	$(0,1,1)$	$-m^2+(48k_0m_0+22k_0+48m_0^2+54m_0+13)m-576k_0^2m_0^2-480k_0^2m_0-97k_0^2-1152k_0m_0^3-1776k_0m_0^2-820k_0m_0-112k_0-576m_0^4-1296m_0^3-1009m_0^2-314m_0-32$
3.2.3	$(0,2,1)$	$-m^2+(48k_0m_0+34k_0+48m_0^2+78m_0+30)m-576k_0^2m_0^2-768k_0^2m_0-253k_0^2-1152k_0m_0^3-2640k_0m_0^2-1936k_0m_0-455k_0-576m_0^4-1872m_0^3-2209m_0^2-1119m_0-205$
3.2.4	$(0,3,1)$	$-m^2+(48k_0m_0+46k_0+48m_0^2+102m_0+52)m-576k_0^2m_0^2-1056k_0^2m_0-481k_0^2-1152k_0m_0^3-3504k_0m_0^2-3460k_0m_0-1110k_0-576m_0^4-2448m_0^3-3817m_0^2-2583m_0-639$
3.2.5	$(1,0,1)$	$-m^2+(48k_0m_0+10k_0+48m_0^2+30m_0+1)m-576k_0^2m_0^2-192k_0^2m_0-13k_0^2-1152k_0m_0^3-912k_0m_0^2-136k_0m_0+2k_0-576m_0^4-720m_0^3-241m_0^2-9m_0+1$
3.2.6	$(1,1,1)$	$-m^2+(48k_0m_0+22k_0+48m_0^2+54m_0+12)m-576k_0^2m_0^2-480k_0^2m_0-97k_0^2-1152k_0m_0^3-1776k_0m_0^2-820k_0m_0-113k_0-576m_0^4-1296m_0^3-1009m_0^2-316m_0-34$
3.2.7	$(1,2,1)$	$-m^2+(48k_0m_0+34k_0+48m_0^2+78m_0+28)m-576k_0^2m_0^2-768k_0^2m_0-253k_0^2-1152k_0m_0^3-2640k_0m_0^2-1912k_0m_0-438k_0-576m_0^4-1872m_0^3-2185m_0^2-1078m_0-190$
3.2.8	$(1,3,1)$	$-m^2+(48k_0m_0+46k_0+48m_0^2+102m_0+51)m-576k_0^2m_0^2-1056k_0^2m_0-481k_0^2-1152k_0m_0^3-3504k_0m_0^2-3460k_0m_0-1111k_0-576m_0^4-2448m_0^3-3817m_0^2-2585m_0-642$
3.2.9	$(2,0,1)$	$-m^2+(48k_0m_0+10k_0+48m_0^2+30m_0+2)m-576k_0^2m_0^2-192k_0^2m_0-13k_0^2-1152k_0m_0^3-912k_0m_0^2-136k_0m_0+2k_0-576m_0^4-720m_0^3-241m_0^2-11m_0+2$
3.2.10	$(2,1,1)$	$-m^2+(48k_0m_0+22k_0+48m_0^2+54m_0+12)m-576k_0^2m_0^2-480k_0^2m_0-97k_0^2-1152k_0m_0^3-1776k_0m_0^2-796k_0m_0-101k_0-576m_0^4-1296m_0^3-985m_0^2-287m_0-25$
3.2.11	$(2,2,1)$	$-m^2+(48k_0m_0+34k_0+48m_0^2+78m_0+29)m-576k_0^2m_0^2-768k_0^2m_0-253k_0^2-1152k_0m_0^3-2640k_0m_0^2-1912k_0m_0-438k_0-576m_0^4-1872m_0^3-2185m_0^2-1080m_0-189$
3.2.12	$(2,3,1)$	$-m^2+(48k_0m_0+46k_0+48m_0^2+102m_0+51)m-576k_0^2m_0^2-1056k_0^2m_0-481k_0^2-1152k_0m_0^3-3504k_0m_0^2-3436k_0m_0-1087k_0-576m_0^4-2448m_0^3-3793m_0^2-2532m_0-613$
3.2.13	$(3,0,1)$	$-m^2+(48k_0m_0+10k_0+48m_0^2+30m_0)m-576k_0^2m_0^2-192k_0^2m_0-13k_0^2-1152k_0m_0^3-912k_0m_0^2-112k_0m_0+7k_0-576m_0^4-720m_0^3-217m_0^2+6m_0+1$
3.2.14	$(3,1,1)$	$-m^2+(48k_0m_0+22k_0+48m_0^2+54m_0+11)m-576k_0^2m_0^2-480k_0^2m_0-97k_0^2-1152k_0m_0^3-1776k_0m_0^2-796k_0m_0-102k_0-576m_0^4-1296m_0^3-985m_0^2-289m_0-28$
3.2.15	$(3,2,1)$	$-m^2+(48k_0m_0+34k_0+48m_0^2+78m_0+27)m-576k_0^2m_0^2-768k_0^2m_0-253k_0^2-1152k_0m_0^3-2640k_0m_0^2-1888k_0m_0-421k_0-576m_0^4-1872m_0^3-2161m_0^2-1039m_0-176$
3.2.16	$(3,3,1)$	$-m^2+(48k_0m_0+46k_0+48m_0^2+102m_0+50)m-576k_0^2m_0^2-1056k_0^2m_0-481k_0^2-1152k_0m_0^3-3504k_0m_0^2-3436k_0m_0-1088k_0-576m_0^4-2448m_0^3-3793m_0^2-2534m_0-617$
4.2.1	$(0,0,1)$	$-4m_0m+48k_0^2m_0+12k_0^2+144k_0m_0^2+82k_0m_0+12k_0+96m_0^3+96m_0^2+25m_0+3$
4.2.2	$(0,1,1)$	$(-4m_0-2)m48k_0^2m_0+24k_0^2+144k_0m_0^2+178k_0m_0+53k_0+96m_0^3+192m_0^2++121m_0+24$
4.2.3	$(0,2,1)$	$(-4m_0-2)m+48k_0^2m_0+36k_0^2+144k_0m_0^2+226k_0m_0+89k_0+96m_0^3+240m_0^2+193m_0+51$
4.2.4	$(0,3,1)$	$(-4m_0-4)m+48k_0^2m_0+48k_0^2+144k_0m_0^2+322k_0m_0+178k_0+96m_0^3+336m_0^2+385m_0+145$
4.2.5	$(1,0,1)$	$(-4m_0-2)m+48k_0^2m_0+12k_0^2+144k_0m_0^2+106k_0m_0+17k_0+96m_0^3+120m_0^2+42m_0+3$
4.2.6	$(1,1,1)$	$(-4m_0-2)m+48k_0^2m_0+24k_0^2+144k_0m_0^2+154k_0m_0+41k_0+96m_0^3+168m_0^2+90m_0+15$
4.2.7	$(1,2,1)$	$(-4m_0-4)m+48k_0^2m_0+36k_0^2+144k_0m_0^2+250k_0m_0+106k_0+96m_0^3+264m_0^2+234m_0+66$
4.2.8	$(1,3,1)$	$(-4m_0-4)m+48k_0^2m_0+48k_0^2+144k_0m_0^2+298k_0m_0+154k_0+96m_0^3+312m_0^2+330m_0+114$
4.2.9	$(2,0,1)$	$-4m_0m+48k_0^2m_0+12k_0^2+144k_0m_0^2+82k_0m_0+12k_0+96m_0^3+96m_0^2+23m_0+3$
4.2.10	$(2,1,1)$	$(-4m_0-2)m+48k_0^2m_0+24k_0^2+144k_0m_0^2+178k_0m_0+53k_0+96m_0^3+192m_0^2+119m_0+24$
4.2.11	$(2,2,1)$	$(-4m_0-2)m+48k_0^2m_0+36k_0^2+144k_0m_0^2+226k_0m_0+89k_0+96m_0^3+240m_0^2+191m_0+51$
4.2.12	$(2,3,1)$	$(-4m_0-4)m+48k_0^2m_0+48k_0^2+144k_0m_0^2+322k_0m_0+178k_0+96m_0^3+336m_0^2+383m_0+143$
4.2.13	$(3,0,1)$	$(-4m_0-2)m+48k_0^2m_0+12k_0^2+144k_0m_0^2+106k_0m_0+17k_0+96m_0^3+120m_0^2+40m_0+2$

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Table 1 – continued from previous page

Case	(n, n_0, k)	$C(n)$
4.2.14	$(3,1,1)$	$(-4m_0-2)m+48k_0^2m_0+24k_0^2+144k_0m_0^2+154k_0m_0+41k_0+96m_0^3+168m_0^2+88m_0+14$
4.2.15	$(3,2,1)$	$(-4m_0-4)m+48k_0^2m_0+36k_0^2+144k_0m_0^2+250k_0m_0+106k_0+96m_0^3+264m_0^2+232m_0+64$
4.2.16	$(3,3,1)$	$(-4m_0-4)m+48k_0^2m_0+48k_0^2+144k_0m_0^2+298k_0m_0+154k_0+96m_0^3+312m_0^2+328m_0+112$

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